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Purposefully different,

consistently excellent

Welcome to EEN1037: Web Applications Development

Who am I?

- Joe Mullally
  - Research Assistant
  - Edge-Computing project between the
    School of Electronic Engineering
    School of Law & Government

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- Research Interests
  - Internet of Things (IoT)
  - Environmental Sensors
  - Containers (Docker, etc)
  - Data Security
  - Web Services & APIs
  - Databases

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Previously EE417

- Dr. Ali Intizar
  - Assistant Professor
  - School of Electronic Engineering
  http://intizarali.org/

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- Research Interests
  - Internet of Things (IoT)
  - Web Data Integration
  - Semantic Web
  - Data Analytics and AI
  - IoT Applications

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Course Material

- Slides and Lecture Notes will be provided via Loop
- Zoom Recordings
- Primary reference
  - W3Schools Tutorial
  - Mozilla MDN Documentation
  - Django Project Documentation
  - Online Reference Material
  - No textbook recommended
  - Links will be posted or linked via LOOP as needed.

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Lectures

- 12 X Lectures
- Every Monday 14:00 – 17:00
- 2 hours lecture in Lecture Hall SA106
- 1-hour hands-on session in computer lab S143

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DCU Loop

- Module contents and learning material will be made available online on the Loop
- Slides
- Recorded videos
- Additional online material
- Module contents will be released on weekly basis

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Discussion Forums

- Two Discussion Forums on Loop
- General Discussion
  - Any question or topic you wish to share and discuss with your peers
    - General information
- Technical Issues
  - problems with using a technology
  - Software installation issues
  - Coding errors
- Please post any general inquiry or technical issue on the relevant forum rather than sending individual emails
- Netiquettes
  - Don't post a new topic every time but rather look first and reply to an existing thread.
  - Please strictly adhere to netiquettes https://www.Rasmussen.edu/student-experience/college-life/netiquette-guidelines-every-online-student-needs-to-know/
  - I will moderate and respond to questions on a weekly basis

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Course Contents

(indicative and not necessarily in the same order)

- **Web Application Design and Architecture**
  - Introduction to web applications and web programming including the underlying architectures, technologies and protocols for web applications.
- **Client Side Programming:**
  - Introduction to client-side programming for web applications including HTML, CSS (Bootstrap / Tailwind), Javascript and Jquery.
  - Brief introduction to Single Page Applications (SPA) using Vue.js.
- **Server Side Programming**
  - Server-side web application development using Python & Django. Hosting web applications using Nginx webserver in a Docker container.

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Course Contents

(indicative and not necessarily in the same order)

- **Relational Database Systems for Web Applications**
  - Designing databases for Web applications, SQL Query language, database connectivity from the web applications.
- **Web Application Development Frameworks**
  - Introduction to various frameworks for web application development and tools for code testing and maintenance.
- **Web Application Deployment and Maintenance**
  - Introduction to cloud computing and deployment of web applications on the cloud. Maintaining scalability of web applications on the cloud.

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Course Contents

(indicative and not necessarily in the same order)

- **AI**
  - Mass production of AI and what that means for modern web development.
  - Long-term job prospects for those who **understand** engineering fundamentals AND can **execute** project tasks effectively.

DO:

- Learn effectively by reading and going through tutorials.
- Use AI for advice and examples.

DONT:

- Submit AI slop as coursework (marks will be deducted).
- Copy & Paste huge chunks of AI code without understanding it.

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Assessment Plan

- Continuous Assessment (100%)
  - Individual Web Application
    - 4 individual assignments to build a Web application (15+15+15+15)
    - You will select your web application for the assignment 1 submission which can't be changed for subsequent assignments.
    - Gradually enhance the selected Web application to build each assignment on top of each other
    - Total worth 60%
  - Group Project
    - 5 to10 students' teams to work on a complete Web application
    - Groups will be randomly created later in the semester
    - Individual role for each team member (20 marks)
    - Collective group assessment (20 marks)
    - Student who has limited contribution in the group project will not be entitled of group assessment marks
    - Total worth 40%

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- For any further questions
  - Drop me an email at: joseph.mullally@dcu.ie

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